Curriculum Vitae

Jiaqi Bi

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A. IDENTIFICATION

Name: Jiaqi Bi
Email: jbi23@uwo.ca
Website: https://jiaqi-bi-not-phd.github.io/
GitHub: https://github.com/Jiaqi-Bi-Not-PhD
Current Program: PhD in Biostatistics, Schulich School of Medicine & Dentistry, University of Western Ontario
Location: London, ON, Canada

B. EDUCATION

Graduate

- Institution: Schulich School of Medicine & Dentistry, University of Western Ontario
- Degree: PhD in Biostatistics
- Year: Expected Aug. 2028
- Thesis: *Statistical Methods in Two-Phase Sampling for Next Generation Sequencing Data*
- Supervisors: Dr. Osvaldo Espin-Garcia and Dr. Yun-Hee Choi
- Comprehensive Exam Passed: Apr. 30, 2025
- Institution: Schulich School of Medicine & Dentistry, University of Western Ontario
- Degree: MSc in Biostatistics
- Year: Dec. 2024
- Thesis: *Frailty Model with Missing Covariates in Family-Based Studies*
- Supervisors: Dr. Yun-Hee Choi and Dr. Osvaldo Espin-Garcia
- Coursework: Foundations of Epidemiology; Biostatistical Research Methods (A); Survival Analysis; Multivariable Biostatistics (A); Statistical Inference (A); Advanced Data Analysis (A); Generalized Linear Model (A); Multilevel Modelling (PASS/FAIL: PASS); Missing Data (PASS/FAIL: PASS); Advanced Statistical Computing (A); Advanced Regression (A)

Undergraduate

- Institution: Faculty of Arts & Science, University of Toronto
- Degree: H.BSc in Mathematics and Statistics
- Year: Jun. 2022
- Related Coursework: Statistical Methods in Emerging Infectious Disease (A-); Applied Statistical Methods (A+); Methods of Data Analysis I and II (A- for both courses); Time Series (A-); Statistical Machine Learning (A); Statistical Design & Analysis of Experiment (A); Complex Variables; PDE (A+); Combinatorics (A); Groups & Symmetry (A+, Top 1 Student); Number Theory (A+); Geometrics (A+)

C. RESEARCH EXPERIENCE

Research Assistant, Dr. Egide Kalisa Lab, University of Western Ontario, London, Canada — Dec 2024–Present

- Assisting in statistical analyses for environmental projects.
- Providing statistical and technical support to co-author articles.

Graduate Research Fellow, Dr. Yun-Hee Choi and Dr. Osvaldo Espin-Garcia, University of Western Ontario, London, Canada — Jul 2023–Present

- Investigating methods for high-dimensional auxiliary covariates in two-phase sampling in next generation sequencing data.
- Developing novel methods for missing data in familial studies.

Research Assistant, Dr. Marc O. Martel Lab, McGill University, Montreal, Canada (Remote) — Jul 2023–Present

- Conducting statistical analyses in R for multiple lab research projects.
- Collaborating with research teams to study pain outcomes using statistical methods and analyses.
- Co-authoring manuscripts for peer-reviewed publication.

Biostatistician, Henan Provincial People's Hospital, Zhengzhou University, Zhengzhou, China (Remote) — Jul 2023–May 2025

- Performed statistical design and analysis for medical research projects.
- Assisted with publications focused on statistical methods.

Undergraduate Research Assistant, Intelligent Adaptive Intervention Lab, University of Toronto, Toronto, Canada — May 2022–Sep 2022

- Investigated COVID-19 vaccine hesitancy in North America with the research team.
- Conducted statistical analyses in R for survey data involving more than 300 respondents.

D. PUBLICATIONS AND ABSTRACTS

D1. Journal Publications and Accepted Papers

1. **Bi, J.**, Jamison, R. N., Ross, E. L., Edwards, R. R., & Martel, M. O. (2026). Daily perceived change among patients with chronic pain: influence of sociodemographic, clinical, and psychological characteristics. *PAIN Reports, 11*(4), e1449. https://doi.org/10.1097/PR9.0000000000001449
2. Bruneau, A., Sirois, A., Spiegler, G., **Bi, J.**, Petkau, S., Djoudi, Y.-L., Ware, J., Alghamdi, R., Nipa, N., Perreault, M., Pagé, M. G., Ware, M. A., Perez, J., & Martel, M. O. (2026). Factors associated with opioid craving, opioid intake, and opioid misuse in patients with chronic pain: insights from an ecological momentary assessment study. *PAIN, 167*(1), 120–132. https://doi.org/10.1097/j.pain.0000000000003753
3. Chang, E., Chen, F., Wu, Y., Zhang, Y., & **Bi, J.** (2026). Exploring the causal relationship between gut microbiota and pan-cancer: A meta-analysis of Mendelian randomization studies. *Scientific Reports* (Accepted).
4. Chang, E., Wu, L., Li, X., Zhou, J., Zhi, H., Sun, M., Chen, G., **Bi, J.**, Li, L., Li, T., Ma, D., & Zhang, J. (2025). Dexmedetomidine decreases cerebral hyperperfusion incidence following carotid stenting: A double-blind, randomized controlled trial. *Med, 6*(3), 100523. https://doi.org/10.1016/j.medj.2024.09.012
5. Kalisa, E., Archer, S., **Bi, J.**, Lee, K., & Lacap-Bugler, D. (2026). The effect of sampling duration and filter extraction fraction on the measurement of atmospheric particulate-bound polycyclic aromatic hydrocarbons and their nitro-derivatives. *Environmental Research Communications, 8*(5), 055030. https://doi.org/10.1088/2515-7620/ae69a7
6. Kalisa, E., Lacap-Bugler, D., Adams, M., **Bi, J.**, Nsabimana, A., Habiyaremye, G., Uwizeye, G., Lawrence, T., Lee, K., Hayakawa, K., Pointing, S., & Archer, S. D. J. (2026). Variability of airborne microbial communities and associations with organic pollutants in African air particulate matter across land-use types. *Atmospheric Environment, 367*, 121750. https://doi.org/10.1016/j.atmosenv.2025.121750
7. Zhang, T., Liu, J., Dilimulati, H., Yao, S., **Bi, J.**, Yang, Q., Wang, W., Shi, H., Xing, W., Liu, X., Zhang, Y., Chen, J., Ge, J., Springer, M. L., & An, S. (2026). Apelin overexpression remodels aged mesenchymal stem cells to generate miR-486a-5p-enriched exosomes that drive cardiac repair. *Stem Cell Research & Therapy* (Accepted).

D2. Submitted Manuscripts and Manuscripts Under Review

1. Bruneau, A., Sirois, A., **Bi, J.**, Spiegler, G., Perez-Medina-Carballo, R., Babiloni, A. H., Ware, J., Ware, M. A., Perez, J., Lupien, S., & Martel, M. O. (2025). Hypothalamic-pituitary-adrenal (HPA) axis activity, psychological states, and opioid-related problems among patients with chronic pain using opioids: Preliminary investigation using ecological momentary assessment (EMA) methods. *Psychiatry Research* (Under Review).
2. Chang, E., Wu, L., Zhang, T., Chen, Q., Zhou, S., Zhao, H., Howkins, A., **Bi, J.**, Li, X., Zhang, Y., Pan, L., & Ma, D. (2026). ClpP promotes kidney cancer cell malignancy through enhanced mitophagy and decreased ferroptosis. *International Journal of Biological Sciences* (Submitted).
3. Kalisa, E., Adams, M., **Bi, J.**, Nsabimana, A., Habiyaremya, G., Uwizeye, G., Lawrence, T., Lee, K., Nagato, E., Hayakawa, K., Pointing, S., Archer, S., & Lacap-Bugler, D. (2025). Microbial and organic pollutant co-exposure in African air particulate matter: A case for integrated risk assessment. *Environment International* (Under Review).
4. Kalisa, E., Mastin, J., Saini, A., **Bi, J.**, Lee, K., Luginaah, I., Pointing, S., Schuster, J., & Harner, T. (2026). Evaluating bioaerosol monitoring approaches for detecting wildfire-associated microbial signals during the 2023 Canadian wildfire season. *ACS Omega* (Submitted).
5. Martel, M. O., **Bi, J.**, Campbell, C. M., Carriere, J. S., Paschali, M., Lazaridou, A., Meints, S. M., Pester, B., Finan, P. H., Haythornthwaite, J. A., Smith, M. T., & Edwards, R. R. (2025). Daily pain symptoms in patients with knee osteoarthritis: Links to psychological factors, pain sensitivity, and endogenous pain modulation. *Arthritis Care & Research* (Under Revision).

D3. Publications in Final Preparations

1. **Bi, J.**, Sun, J., Dou, H., Yuan, J., Xian, C., & Espin-Garcia, O. (2026). Uncertainty aware prediction from incomplete clinical records: A conformal prediction study of Parkinsonian classification. *In Final Preparation*.
2. **Bi, J.**, Briollais, L., Espin-Garcia, O., & Choi, Y. (2026). Correlated frailty model with missing continuous covariate in family-based study. *In Progress*.
3. **Bi, J.**, Briollais, L., Espin-Garcia, O., & Choi, Y. (2026). Correlated frailty model and penetrance estimates with missing covariates in family-based study: A maximum likelihood multiple imputation approach. *In Progress*.

D4. Accepted Abstracts

1. **Bi, J.**, Sun, J., Dou, H., & Yuan, J. (2026). Class imbalance aware ensemble learning for clinical differentiation of Parkinson disease and atypical parkinsonism. *Statistical Society of Canada Annual Meeting, Case Study #1*.

2. **Bi, J.**, Briollais, L., Espin-Garcia, O., & Choi, Y. (2026). Correlated frailty model with missing continuous covariate in family-based study. *Canadian Statistics Student Conference Meeting*.
3. **Bi, J.**, Espin-Garcia, O., & Choi, Y. (2024). Correlated shared frailty model incorporating ascertainment correction with missing covariates in family-based studies. *Statistical Society of Canada Annual Meeting*.
4. Frankel, L., Sirois, A., Rosenthal, D., Mesli, N., **Bi, J.**, Cui, L., Ware, M., Perez, J., Fitzcharles, M. A., & Martel, M. O. (2024). Exploring the factors contributing to daily cannabis craving and cannabis use among chronic pain patients. *Department of Psychology Annual Research Day, McGill University*.
5. Jamison, R. N., **Bi, J.**, Meints, S., Edwards, R. R., & Martel, M. O. (2024). Perceived improvement or worsening among persons with chronic pain: analyses of pain app data. *International Association for the Study of Pain Meeting*.
6. Martel, M. O., Bruneau, A., Sirois, A., **Bi, J.**, Spiegler, G., Ware, J., Alghamdi, R., Nipa, N., Pagé, G., Ware, M., & Perez, J. (2025). Day-to-day opioid intake among patients with chronic pain using short-acting opioids: Exploring the role of pain intensity, psychological states, and physical dependence. *Canadian Pain Society Meeting*.
7. Sirois, A., Frankel, L., Spiegler, G., **Bi, J.**, Rosenthal, D., Verner, M., Binshihah, A., Hudon, J., Ben-Sasson, M., Pagé, G., Perez, J., Ware, M., Fitzcharles, M.-A., & Martel, M. O. (2025). Daily determinants of cannabis craving and cannabis intake among persons with chronic pain: Preliminary insights from an ecological momentary assessment (EMA) study. *Canadian Pain Society Meeting*.
8. Sirois, A., Verner, M., Bruneau, A., **Bi, J.**, Perez, J., Shir, Y., & Martel, M. O. (2024). Week-to-week reports of physical and cognitive side effects associated with opioids: Insights from a longitudinal study among persons with chronic pain. *Canadian Pain Society Meeting*.

D5. Presentations

1. **Bi, J.**, Sun, J., Dou, H., & Yuan, J. "Class Imbalance Aware Ensemble Learning for Clinical Differentiation of Parkinson Disease and Atypical Parkinsonism." *Statistical Society of Canada Annual Meeting, Case Study #1* (Team Poster Presentation), Hamilton, Canada. — Jun 2026
2. **Bi, J.** "Correlated Frailty Model with Missing Continuous Covariate in Family-Based Study: A Maximum Likelihood Multiple Imputation Approach." *Canadian Statistics Student Conference Meeting* (Poster Presentation), Hamilton, Canada. — May 2026
3. **Bi, J.** "Correlated Shared Frailty Model Incorporating Ascertainment Correction with Missing Covariates in Family-Based Studies." *Statistical Society of Canada Annual Meeting*, St. John's, Canada. — Jun 2024
4. **Bi, J.** "Correlated Shared Frailty Model Incorporating Ascertainment Correction with Missing Covariates in Family-Based Studies." *Epidemiology and Biostatistics Research Day, University of Western Ontario* (Oral Presentation), London, Canada. — Apr 2024
5. **Bi, J.** "Factors Associated with Day-to-Day Pain Fluctuations and Pain Flares Among Patients with Chronic Pain." *McGill CERC Pain & Genetics Research Day*, Montreal, Canada. — Oct 2023

D6. Software

1. **Bi, J.** (n.d.). *ggmultilevel* (Version 1.0.0) [Computer software]. GitHub. https://github.com/Jiaqi-Bi-Not-PhD/ggmultilevel

E. TEACHING

Teaching Assistantships

BIOSTATS3100A/9509 – Biostatistical Methods for Epidemiology, Teaching Assistant, University of Western Ontario — Sep 2026–Present

- Supporting lab tutorials, including SAS instruction.
- Course instructor: Dr. Yun-Hee Choi

BIOSTATS3100A/9509 – Biostatistical Methods for Epidemiology, Teaching Assistant, University of Western Ontario — Sep 2025–Dec 2025

- Instructed weekly lab tutorials, held office hours, proctored exams, graded assessments, and supported course material preparation.
- Course instructor: Dr. Yun-Hee Choi

MPH9017B – Health Informatics, Teaching Assistant, University of Western Ontario — Jan 2025–May 2025

- Held office hours, proctored exams, graded assessments, and supported course material preparation.
- Course instructor: Dr. Steve Lee

MPH9017B – Health Informatics, Teaching Assistant, University of Western Ontario — Jan 2025–May 2025

- Held office hours, proctored exams, graded assessments, and supported course material preparation.
- Course instructor: Dr. Dan Lizotte

BIOSTATS3110B – Multivariable Methods in Biostatistics, Teaching Assistant, University of Western Ontario — Jan 2025–May 2025

- Held office hours, proctored exams, graded assessments, and supported course material preparation.
- Course instructor: Dr. Guangyong Zou

BIOSTATS3110B – Multivariable Methods in Biostatistics, Teaching Assistant, University of Western Ontario — Jan 2024–May 2024

- Instructed weekly lab tutorials, held office hours, proctored exams, graded assessments, and supported course material preparation.
- Course instructor: Dr. Osvaldo Espin-Garcia
- This course received the Teaching Excellence Award in 2024.

F. HONORS, CERTIFICATIONS, AND SKILLS

F1. Awards

Graduate Student Enrichment Scholarship, Canadian Statistical Sciences Institute (CANSSI) — Apr 2025–Apr 2026

$15,000 awarded by merit.

Western Graduate Research Scholarship, University of Western Ontario — Sep 2022–Aug 2027

$10,800 per year for 2 years awarded by the department for Master's study.

$4,500 per year for 4 years awarded by the department for Doctoral study.

WEBSC Travel Award — 2026

$500 travel award.

Conference Travel Awards, Canadian Statistics Student Conference & CANSSI — June 2026

$100 Student travel award.

Awarded based on merit, submitted abstract score ranked #1 among Ontario submissions.

Conference Travel Awards, University of Western Ontario — Aug 2024

$300 travel award.

F2. Certifications

- SAS Certified Specialist: Base Programming Using SAS 9.4 (2021)
- SAS Certified Professional: Advanced Programming Using SAS 9.4 (2022)

F3. Technical Skills

- Programming: R, Stata, SAS, SQL, Python, Blimp Software
- Tools: LaTeX, R Markdown, R Quarto, Git, Bash/Shell scripting, Unix/Linux systems, HPC, Microsoft Office

G. SERVICE AND OTHER CONTRIBUTIONS

G1. Academic Service

Conference Organization and Student Representation

- Scientific Program Committee Member, Canadian Statistics Student Conference (CSSC) 2027 (2026–2027).
- Equity, Diversity, and Inclusion (EDI) Representative, Epidemiology and Biostatistics Students' Council, University of Western Ontario (2025–2026).

Presentation and Poster Judge

- Western Summer Research Conference Expo (2024)
- Western EpiBio Research Day (2025)

Journal Reviewer

- *Advanced Science* (1 article reviewed)

G2. Other Contributions

- Author of the University of Western Ontario (Schulich School of Medicine & Dentistry) Beamer Template, published on Overleaf.